

List 2 - Fin Ele

1. **Cauchy-Schwarz inequality.** Consider a space V with inner product $(\cdot, \cdot)_V$ and induced norm $\|\cdot\|_V$. For any $v, w \in V$

$$|(v, w)_V| \leq \|v\|_V \|w\|_V \quad (2.29)$$

Prove this important inequality. This inequality is a particular case of the more general Hölder's inequality. Let $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$, $v \in L^p(\Omega)$ and $w \in L^q(\Omega)$, then $vw \in L^1(\Omega)$ and

$$\|vw\|_{L^1(\Omega)} \leq \|v\|_{L^p(\Omega)} \|w\|_{L^q(\Omega)} \quad (2.30)$$

$$\Delta \leq 0: -4(v, w)_V^2 - 4\|v\|_V^2 \|w\|_V^2 \leq 0$$

$$(v, w)_V^2 \leq \|v\|_V^2 \|w\|_V^2 \Rightarrow |(v, w)_V| \leq \|v\|_V \|w\|_V$$

$$\|v\|_V = \sqrt{(v, v)}$$

$$\begin{aligned} (v, w)_V &= (v, w - v + v)_V = (v, w - v)_V + (v, v)_V = (v, w - v)_V + \|v\|_V^2 = \\ &= (v - w + w, w - v)_V + \|v\|_V^2 = (v - w, w - v)_V + (w, w - v)_V + \|v\|_V^2 = \\ &= -\|v - w\|_V^2 + \|w\|_V^2 + \|v\|_V^2 - (w, v)_V \\ \Rightarrow (v, w)_V &= \frac{1}{2} (\|v\|_V^2 - \|v - w\|_V^2 + \|w\|_V^2) \end{aligned}$$

$$(\|v\|_V - \|w\|_V)^2 = \|v\|_V^2 - 2\|v\|_V \|w\|_V + \|w\|_V^2$$

$$\Rightarrow (v, w)_V = \frac{1}{2} \left((\|v\|_V - \|w\|_V)^2 + 2\|v\|_V \|w\|_V - \|v - w\|_V^2 \right)$$

$$\Rightarrow |(v, w)_V| = \frac{1}{2} \left| (\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 + 2\|v\|_V \|w\|_V \right| \leq$$

$$\leq \frac{1}{2} \left| (\|v\|_V - \|w\|_V)^2 - \|v - w\|_V^2 \right| + \|v\|_V \|w\|_V \leq$$

$$\leq \|v\|_V \|w\|_V$$

2. **Triangle inequality.** Using the Cauchy-Schwarz inequality, prove that for any $v, w \in V$

$$\|v + w\|_V \leq \|v\|_V + \|w\|_V \quad (2.31)$$

$$\begin{aligned} \|v + w\|_V^2 &= (v + w, v + w)_V = (v + w, v)_V + (v + w, w)_V \leq \\ &\leq \|v + w\|_V (\|v\|_V + \|w\|_V) \end{aligned}$$

$$\Rightarrow \|v + w\|_V \leq \|v\|_V + \|w\|_V$$

3. Prove that the sequence of Equation 2.12 is a Cauchy sequence.

Consider the space $C^0(\Omega)$ on $\Omega = [0, 1]$, equipped with the $L^2(\Omega)$ norm, $\|v\|_{L^2(\Omega)} = \left(\int_0^1 v^2 dx \right)^{1/2}$, and the sequence of continuous functions $\{v_n\}_{n=1}^\infty$ illustrated in Figure 2.6:

$$v_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{2}, \\ \left(x - \frac{1}{2}\right)^{\frac{1}{n}} & \text{if } \frac{1}{2} \leq x \leq 1, \end{cases} \quad (2.12)$$

$$\|v_n - v_m\|_{L^2(\Omega)} = \left(\int_0^1 (v_n(x) - v_m(x))^2 dx \right)^{1/2} = \left(\int_{1/2}^1 \left(\left(x - \frac{1}{2}\right)^{\frac{1}{n}} - \left(x - \frac{1}{2}\right)^{\frac{1}{m}} \right)^2 dx \right)^{1/2}$$

$$\left(\int_{1/2}^1 \left(x - \frac{1}{2} \right)^{\frac{2}{n}} - 2 \left(x - \frac{1}{2} \right)^{\frac{1}{n} + \frac{1}{m}} + \left(x - \frac{1}{2} \right)^{\frac{2}{m}} dx \right)^{1/2} = (1)$$

Note that

$$\int_{1/2}^1 \left(x - \frac{1}{2} \right)^k dx = \int_0^{1/2} u^k du = \frac{u^{k+1}}{(k+1)} \Big|_0^{1/2} = \frac{1}{2^{k+1}(k+1)}$$

$$u = x - \frac{1}{2}$$

$$du = dx$$

$$\Rightarrow (1) = \frac{1}{2^{\frac{2}{n}+1} \left(\frac{2}{n} + 1 \right)} - \frac{2}{2^{\frac{1}{n} + \frac{1}{m} + 1} \left(1 + \frac{1}{n} + \frac{1}{m} \right)} + \frac{1}{2^{\frac{2}{m}+1} \left(\frac{2}{m} + 1 \right)} =$$

$$\frac{1}{2^{\frac{2}{n}+1} \left(\frac{2}{n} + 1 \right)} + \frac{1}{2^{\frac{2}{m}+1} \left(\frac{2}{m} + 1 \right)} - \frac{1}{2^{\frac{1}{n} + \frac{1}{m} + 1} \left(1 + \frac{1}{n} + \frac{1}{m} \right)} = (2)$$

(Pesso duridosa!!!)

Noting that (2) is continuous as a function of m and n for $m, n \geq 1$, if we take $m, n \rightarrow \infty$

$$(2) \rightarrow \frac{1}{2} + \frac{1}{2} - 1 = 0$$

As the limit exists, $\forall \epsilon > 0$, $\exists N > 0$ such that $n, m > N \Rightarrow$

$$\Rightarrow \|v_n - v_m\|_{L^2(\Omega)} < \epsilon$$

5. Let $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^3$. Write the partial derivative of f corresponding to $\vec{\alpha} = (2, 0, 0)$ and $\vec{\alpha} = (0, 1, 1)$.

$$\vec{\alpha} = (2, 0, 0):$$

$$\vec{\alpha} = (0, 1, 1):$$

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_1^2}$$

$$D^{\vec{\alpha}} f = \frac{\partial^2 f}{\partial x_2 \partial x_3}$$

10. Let Ω be bounded and let $1 \leq p \leq q < \infty$. Prove the inclusion stated without proof in Section 2.3,

$$L^q(\Omega) \subset L^p(\Omega).$$

a) Apply Hölder's inequality, Eq. (2.29), to the product $|f|^p \cdot 1$ with the exponents $r = q/p$ and $r' = r/(r-1)$, and conclude that

$$\|f\|_{L^p(\Omega)} \leq |\Omega|^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q(\Omega)},$$

where $|\Omega|$ denotes the measure of Ω . Observe where the boundedness of Ω was used.

b) Show that the inclusion is strict: take $\Omega = (0, 1)$, $p = 1$, $q = 2$ and $f(x) = x^{-s}$, $s > 0$. Determine for which s one has $f \in L^1(\Omega)$ and for which $f \in L^2(\Omega)$.

c) Show that boundedness of Ω is essential: taking $\Omega = (1, \infty)$ and $f(x) = 1/x$, verify that $f \in L^2(\Omega)$ but $f \notin L^1(\Omega)$.

$$L^p(\Omega) = \left\{ f: \Omega \rightarrow \mathbb{R} \mid \int_{\Omega} |f|^p dx < +\infty \right\}$$

In fact,

$$\|f\|_{L^p(\Omega)} = \left(\int_{\Omega} |f|^p dx \right)^{1/p}$$

$$\frac{1}{r} + \frac{1}{r'} = \frac{1}{r} + \frac{r-1}{r} = \frac{r}{r} = 1$$

$$\|f\|_{L^p(\Omega)} = \int_{\Omega} |f|^p dx = \|f\|_{L^p(\Omega)}^p \quad (1)$$

we can apply Hölder inequality for the exponents

$$\text{And } \|f\|_{L^p(\Omega)} = \| |f|^p \|_{L^1(\Omega)}^{1/p} \stackrel{\text{Hölder}}{\leq} \| |f|^p \|_{L^r(\Omega)}^{1/p} \|1\|_{L^{r'}(\Omega)}^{1/p} =$$

$$= \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}} \left(\int_{\Omega} 1^{r'} dx \right)^{\frac{1}{p}} = \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}} \left(\int_{\Omega} 1 dx \right)^{\frac{1}{p}} =$$

$$= |\Omega|^{\frac{1}{p}} \left(\int_{\Omega} |f|^p dx \right)^{\frac{p}{p}}$$

$$\frac{1}{r'} = \left(\frac{p}{p} - 1 \right) \frac{p}{p} = \frac{p-p}{p} = \frac{p-p}{p} \Rightarrow \| |f|^p \|_{L^1(\Omega)} = |\Omega|^{\frac{p-p}{p}} \|f\|_{L^p(\Omega)}^p$$

Using (1):

$$\|f\|_{L^p(\Omega)} = \| |f|^p \|_{L^1(\Omega)}^{1/p} \leq |\Omega|^{\frac{p-p}{p}} \|f\|_{L^p(\Omega)}^p = |\Omega|^{\frac{1}{p} - \frac{1}{p}} \|f\|_{L^p(\Omega)}^p$$

If Ω is not bounded, then $L^p(\Omega)$ is not finite, which gives us nothing useful, as $\|f\|_{L^p(\Omega)}$ could be ∞ .

This shows that if $f \in L^p(\Omega)$, then $\|f\|_{L^p(\Omega)} < \infty \Rightarrow f \in L^p(\Omega)$.

b) $\Omega = (0,1)$; $p=1$; $q=2$
 $f(x) = x^{-s}$, $s > 0$

1) Finding s such that

$$f \in L^1(\Omega) \Leftrightarrow \int_{\Omega} |f(x)| dx < \infty$$

$$\int_{\Omega} |f(x)| dx = \int_0^1 x^{-s} dx$$

if $s < 1$ $\int_{\Omega} |f(x)| dx = \frac{x^{1-s}}{1-s} \Big|_0^1 = \frac{1^{1-s}}{1-s} - \frac{0^{1-s}}{1-s} = \frac{1}{1-s} < \infty$

if $s > 1$ $\int_{\Omega} |f(x)| dx = \frac{1}{(1-s)x^{s-1}} \Big|_0^1 = \frac{1}{1-s} - \lim_{x \rightarrow 0^+} \frac{1}{(1-s)x^{s-1}} \rightarrow \infty$

if $s = 1$ $\int_{\Omega} |f(x)| dx = \ln(x) \Big|_0^1 = 0 - \lim_{x \rightarrow 0^+} \ln(x) \rightarrow \infty$

$f \in L^1(\Omega)$ when $s < 1$

2) Finding s such that $f \in L^2(\Omega)$

$$f \in L^2(\Omega) \Leftrightarrow \int_{\Omega} |f(x)|^2 dx < \infty$$

$$\int_{\Omega} |f(x)|^2 dx = \int_{\Omega} x^{-2s} dx = \int_{\Omega} x^{-t} dx, \text{ with } t = 2s.$$

We know this converges for $t < 1 \Rightarrow s < \frac{1}{2}$

So, $\forall s \in [\frac{1}{2}, 1)$, $f \in L^1(\Omega)$ but $f \notin L^2(\Omega)$

$\Rightarrow$ For $1 \leq p < q < \infty$, $\exists f \in L^p(\Omega)$ such that $f \notin L^q(\Omega) \Rightarrow L^p(\Omega) \not\subset L^q(\Omega) \Rightarrow L^q(\Omega) \subsetneq L^p(\Omega)$

c) $\Omega = (1, \infty)$ $f(x) = \frac{1}{x}$

$$\int_{\Omega} |f(x)|^p dx = \int_1^{\infty} \frac{1}{x^p} dx = \frac{-1}{x^{p-1}} \Big|_1^{\infty} = -\lim_{x \rightarrow \infty} \frac{1}{x^{p-1}} - \left(-\frac{1}{1}\right) = 0 + 1 = 1 < \infty$$

$\Rightarrow f \in L^p(\Omega)$

$$\int_{\Omega} |f(x)| dx = \int_1^{\infty} \frac{1}{x} dx = \ln(x) \Big|_1^{\infty} = \lim_{x \rightarrow \infty} \ln(x) = \infty$$

$\Rightarrow f \notin L^1(\Omega)$

$\Rightarrow$ The boundedness of Ω is necessary for the inclusion to be valid.

13. Consider again the function $f(x) = 1 - |x|$ on $\Omega = (-1, 1)$ introduced earlier, whose first weak derivative is the piecewise constant function

$$g(x) = \begin{cases} 1, & -1 < x < 0, \\ -1, & 0 < x < 1, \end{cases}$$

We wish to show that the second weak derivative of f does not exist.

a) Suppose that $h = D^2 f$ exists. Repeating on g the integration by parts performed in earlier, splitting the interval at $x = 0$ and recalling that g is constant on each half, show that necessarily

$$\int_{-1}^1 h(x) \varphi(x) dx = -2 \varphi(0) \quad \forall \varphi \in \mathcal{D}(\Omega).$$

Note that, unlike the previous case, the boundary terms at $x = 0$ no longer cancel, because g is discontinuous there. Identify the functional on the right-hand side as $-2\delta_0$, where δ_0 is the Dirac delta, and explain informally why no ordinary function can reproduce it.

b) (Optional for undergrads) Prove it rigorously. Let $\varphi_1 \in \mathcal{D}(-1, 1)$ satisfy $\varphi_1(0) = 1$ and $0 \leq \varphi_1 \leq 1$ (for instance the bump function of Figure 2.8), and set $\varphi_n(x) = \varphi_1(nx)$.

- Verify that $\varphi_n \in \mathcal{D}(\Omega)$ and that $\text{supp } \varphi_n \subset (-1/n, 1/n)$, so that the identity of part (a) gives $\int_{-1}^1 h \varphi_n dx = -2$ for every n .
- Show that, for any $h \in L^1_{\text{loc}}(\Omega)$,

$$\left| \int_{-1}^1 h \varphi_n dx \right| \leq \int_{-1/n}^{1/n} |h| dx \xrightarrow{n \rightarrow \infty} 0,$$

justifying the limit by the absolute continuity of the Lebesgue integral. Conclude that $-2 = 0$, a contradiction, so no such h exists.

Being continuous and piecewise C^1 therefore buys one weak derivative, but not two. Continuity by itself buys nothing: a Weierstrass function is continuous and possesses no weak derivative at all. This is why continuous piecewise linear finite elements belong to $H^1(\Omega)$ but not to $H^2(\Omega)$.

a) Let $\varphi \in C^\infty(\Omega)$ such that $\lim_{x \rightarrow 1^-} \varphi(x) = \lim_{x \rightarrow -1^+} \varphi(x) = 0$

$$\int_{-1}^1 g(x) \varphi'(x) dx = \int_{-1}^0 \varphi'(x) dx - \int_0^1 \varphi'(x) dx = \varphi(x) \Big|_{-1}^0 - \varphi(x) \Big|_0^1 = \varphi(0) - (-\varphi(0)) = 2\varphi(0)$$

Supposing h is a weak derivative for g , we have

$$-\int_{-1}^1 h(x) \varphi(x) dx = \int_{-1}^1 g(x) \varphi'(x) dx = 2\varphi(0) \Rightarrow$$

$$\Rightarrow \int_{-1}^1 h(x) \varphi(x) dx = -2\varphi(0) = -2 \int_{-1}^1 \delta_0(x) \varphi(x) dx$$

As g is differentiable at $x \neq 0$, we must have $h = g'$ for $x \neq 0$. But then, $h(x) \neq 0$ only at $x = 0$. But $\{0\}$ is a set with measure zero. If $h(0)$ is finite, then we'd have $\int_{-1}^1 h(x) \varphi(x) dx = 0$, $\forall \varphi \in \mathcal{D}$, absurd. So, h must be a Dirac's delta.

Dirac's delta.

b) $\varphi_i \in \mathcal{D}(-1, 1) = \{\varphi \in C^\infty(-1, 1) \mid \text{supp } \varphi \text{ is compact in } (-1, 1)\}$
 $\text{supp } \varphi = \{x \in \Omega : \varphi(x) \neq 0\}$

$$\varphi_i(0) = 1; \quad 0 \leq \varphi_i \leq 1$$

$$\varphi_n(x) = \varphi_i(nx)$$

As φ_i is defined only at $(-1, 1)$, $\varphi_n(x) = \varphi_i(nx) \Rightarrow \varphi_n$ is defined at $(-\frac{1}{n}, \frac{1}{n}) = \Omega_n$

$$\text{supp}(\varphi_n) \subset \left(-\frac{1}{n}, \frac{1}{n}\right)$$

take $x_0 \in \text{supp}(\varphi_n)$. $\forall \varepsilon > 0$, $\exists x_1 \in B_\varepsilon(x_0)$ such that $\varphi_n(x_1) \neq 0 \Rightarrow \varphi_1(nx_1) \neq 0 \Rightarrow nx_1 \in (-1, 1) \Rightarrow x_1 \in \left(-\frac{1}{n}, \frac{1}{n}\right)$

$$\text{take } \varepsilon = \frac{1}{2} \min \left\{ \left|x_0 - \frac{1}{n}\right|, \left|x_0 + \frac{1}{n}\right| \right\}$$

Suppose by absurdity $x_0 \geq \frac{1}{n}$

$$|x_0 - x_1| \geq |x_0| - |x_1| \geq \frac{1}{n} - |x_1| \Rightarrow |x_1| \geq \frac{1}{n} - |x_0 - x_1| \geq$$

$$\geq \frac{1}{2} \min \left\{ \left|x_0 - \frac{1}{n}\right|, \left|x_0 + \frac{1}{n}\right| \right\} + \frac{1}{n} = \frac{1}{n} - \frac{1}{2} \left(x_0 - \frac{1}{n}\right) =$$

$$\frac{3}{2n} - \frac{x_0}{2} > \frac{3}{2n} - \frac{1}{2n} = \frac{2}{2n} = \frac{1}{n} \quad \therefore |x_1| \geq \frac{1}{n} \quad \downarrow$$

Analogous for $x_0 < -\frac{1}{n}$

$$\text{So } |x_0| < \frac{1}{n} \Rightarrow \text{supp}(\varphi_n) \subset \left(-\frac{1}{n}, \frac{1}{n}\right)$$

φ_n is continuous, as φ_1 is continuous and $\text{supp}(\varphi_n) \subset \left(-\frac{1}{n}, \frac{1}{n}\right) \subset (-1, 1)$. So $\varphi_n \in \mathcal{D}(-1, 1)$

As $\varphi_n(0) = (\varphi_1(0) = 1$, we then have

$$\int_{-1}^1 h \varphi_n dx = -2$$

ii) We know $0 \leq \varphi_n \leq 1 \Rightarrow$

$$\left| \int_{-1}^1 h \varphi_n dx \right| = \left| \int_{-1/n}^{1/n} h \varphi_n dx \right| \leq \int_{-1/n}^{1/n} |h| \varphi_n dx =$$

$$= \int_{-1/n}^{1/n} |h| \varphi_n dx \leq \int_{-1/n}^{1/n} |h| dx \xrightarrow{n \rightarrow \infty} 0$$

Absolute continuity of the Lebesgue integral

$$\text{But } \int_{-1}^1 h \varphi_n dx = -2 \varphi_n(0) = -2 \Rightarrow \left| \int_{-1}^1 h \varphi_n dx \right| = 2$$

$$\Rightarrow 2 = 0, \text{ absurd.}$$

9. (Optional for undergrads) Show that the set $C_b(\Omega)$ of bounded continuous functions on a domain Ω , equipped with the norm $\|\cdot\|_{L^\infty(\Omega)}$ of Eq. (2.8), is a Banach space. Compare with Eqs. (2.11)–(2.12), where the same space fails to be complete with respect to $\|\cdot\|_{L^2(\Omega)}$. Hint: proceed in three steps. (i) Let $\{f_n\}$ be a Cauchy sequence in $\|\cdot\|_{L^\infty(\Omega)}$ and observe that, for each fixed $x \in \Omega$, the numerical sequence $\{f_n(x)\}$ is Cauchy in $\mathbb{R}$; since $\mathbb{R}$ is complete, this defines a candidate limit $f(x)$. (ii) Show that $\|f_n - f\|_{L^\infty(\Omega)} \rightarrow 0$, i.e. that the convergence is uniform and not merely pointwise: given $\varepsilon > 0$, take N such that $|f_n(x) - f_m(x)| \leq \varepsilon$ for all $n, m \geq N$ and all $x \in \Omega$; fix $n \geq N$ and let $m \rightarrow \infty$, which gives $|f_n(x) - f(x)| \leq \varepsilon$ for every $x \in \Omega$. Note that N does not depend on x . (iii) Show that f is continuous, by the $\varepsilon/3$ argument

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)|,$$

choosing first n large (using step (ii)) and then $|x - y|$ small (using the continuity of that fixed f_n).

$$C_b(\Omega) = \{f: \Omega \rightarrow \mathbb{R} : f \text{ cont. limited}\}$$

$$\exists N \text{ st } n > N \Rightarrow$$

i) Take $\varepsilon > 0$ and $x \in \Omega$. As f_n is Cauchy with $L^\infty(\Omega)$
 $\Rightarrow |f_n(x) - f_m(x)| \leq \sup_{y \in \Omega} |f_n(y) - f_m(y)| = \|f_n - f_m\|_{L^\infty(\Omega)} < \varepsilon$

$\Rightarrow \exists f: \Omega \rightarrow \mathbb{R}$, $f \in C_b(\Omega)$ such that $f_n \rightarrow f$ pointwise

ii) $|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_{L^\infty(\Omega)} < \varepsilon$, $\forall n, m > N$, $\forall x \in \Omega$
 fixed n , $m \rightarrow \infty$

$$|f_n(x) - f(x)| < \varepsilon, \forall x \in \Omega$$

iii) Using the continuity of f_n : Fix $x \in \Omega$. Take $\varepsilon > 0$. If $|x - y| < \delta$ and $n > N$

$$|f(x) - f(y)| = |f(x) - f_n(x) + f_n(x) - f_n(y) + f_n(y) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$$

$\Rightarrow f$ is continuous

$$f(x) = f(x) - f_n(x) + f_n(x)$$

$$\Rightarrow |f(x)| = |f(x) - f_n(x) + f_n(x)| \leq |f(x) - f_n(x)| + |f_n(x)| < \varepsilon + M$$

$\Rightarrow f$ is bounded

$\Rightarrow f \in C_b(\Omega) \Rightarrow C_b(\Omega)$ is complete with $L^\infty(\Omega)$

Formolizer

4. Consider the space $H^1(\Omega)$, $\Omega = (a, b)$, the Hilbert space of functions $v: (a, b) \rightarrow \mathbb{R}$ and the corresponding induced norm of Equation 2.15, i.e., $\|v\|_{H^1(a,b)} = \left(\int_a^b (v^2 + v'^2) dx \right)^{\frac{1}{2}}$. Show that

$$\|v\|_{H^1(a,b)} = \left(\|v\|_{L^2(\Omega)}^2 + \|v'\|_{L^2(\Omega)}^2 \right)^{\frac{1}{2}}$$

$$\|v\|_{H^1(a,b)} = \|v\|_{L^2(\Omega)} + \|\kappa(x)v'\|_{L^2(\Omega)}$$

where $0 \leq \kappa_{\min} < \kappa(x) \leq \kappa_{\max}$, is an equivalent norm to $\|\cdot\|_{H^1(a,b)}$.

$$v: \Omega \rightarrow \mathbb{R}^d$$

$$(v, w)_{H^1(\Omega)} = \int_{\Omega} (v \cdot w + \nabla v : \nabla w) dx$$

$$\nabla v : \nabla w = \sum_{i,j=1}^d (\nabla v)_{ij} (\nabla w)_{ij}$$

$$\|v\|_{H^1(a,b)} = \left(\int_a^b v^2 + v'^2 dx \right)^{\frac{1}{2}} = \left(\int_a^b v^2 dx + \int_a^b v'^2 dx \right)^{\frac{1}{2}} =$$

$$= \left(\|v\|_{L^2(a,b)}^2 + \|v'\|_{L^2(a,b)}^2 \right)^{\frac{1}{2}} = \|v\|_{H^1(a,b)}$$

$$\|v\|_{H^1(a,b)} = \left(\|v\|_{L^2(a,b)}^2 + \|\kappa v'\|_{L^2(a,b)}^2 \right)^{\frac{1}{2}} = \left(\int_a^b v^2 dx + \int_a^b \kappa^2 v'^2 dx \right)^{\frac{1}{2}}$$

6. Let $f: \Omega \rightarrow \mathbb{R}, \Omega \subset \mathbb{R}^2$. Write in detail the definition of the space $H^2(\Omega)$ and its corresponding norm (Equation 2.27).

Definition 2.4.2 (Sobolev Space) For $m \in \mathbb{N}_0$ and $1 \leq p \leq \infty$, the Sobolev space $W^{m,p}(\Omega)$ is defined as

$$W^{m,p}(\Omega) = \left\{ f \in L^p(\Omega) : D^{\vec{\alpha}} f \in L^p(\Omega) \quad \forall \vec{\alpha} \text{ with } |\vec{\alpha}| \leq m \right\}, \quad (2.26)$$

equipped with the norm

$$\|f\|_{W^{m,p}(\Omega)} = \left(\sum_{|\vec{\alpha}| \leq m} \|D^{\vec{\alpha}} f\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}. \quad (2.27)$$

$$H^2(\Omega) = W^{2,2}(\Omega) = \left\{ f \in L^2(\Omega) : D^{\vec{\alpha}} f \in L^2(\Omega), \right.$$

$$\left. \forall \vec{\alpha} \text{ with } |\vec{\alpha}| \leq 2 \right\}$$

$$\|f\|_{H^2(\Omega)} = \|f\|_{W^{2,2}(\Omega)} = \left(\sum_{|\vec{\alpha}| \leq 2} \|D^{\vec{\alpha}} f\|_{L^2(\Omega)}^2 \right)^{\frac{1}{2}}$$

7. (Optional) Show that the space

$$V = \left\{ w : (0,1) \rightarrow \mathbb{R}, \int_0^1 w(x)^2 dx < +\infty, \int_0^1 w'(x)^2 dx < +\infty \right\}$$

is contained in $C^0(0,1)$.

Let $w \in V$. Suppose $w \notin C^0(0,1)$ by contradiction.

So $\exists x_0 \in (0,1)$ such that w is discontinuous at x_0 . Therefore, $\exists \varepsilon > 0$ such that $\forall \delta > 0, \exists x \in B_\delta(x_0)$ st. $|w(x) - w(x_0)| \geq \varepsilon$

Let $\delta_n \rightarrow 0, \delta_n > 0$. For every n , take $a_n \in B_{\delta_n}(x_0)$ st. $|w(a_n) - w(x_0)| \geq \varepsilon$. Without loss of generality, assume $a_n < x_0$

$$w(x_0) - w(a_n) = \int_{a_n}^{x_0} w' dx \leq \int_{a_n}^{x_0} |w'| dx = \|w'\|_{L^1(\Omega)}$$

$$\text{Note that } \int_{a_n}^{x_0} |w'| dx < \infty \Rightarrow \int_{a_n}^{x_0} |w'| dx < \infty, \text{ as}$$

$(0,1)$ is bounded

$$\text{Let } w_n = \sup_{x \in (a_n, x_0)} |w'(x)|$$

$$\int_{a_n}^{x_0} |w'| dx \leq |x_0 - a_n| w_n$$

$$w_n \text{ is a decreasing sequence and } |x_0 - a_n| \rightarrow 0 \Rightarrow |w(x_0) - w(a_n)| = \left| \int_{a_n}^{x_0} w' dx \right| = \int_{a_n}^{x_0} |w'| dx \rightarrow 0$$

absurd, because $|w(x_0) - w(a_n)| \geq \varepsilon$

8. Young's inequality, or arithmetic-geometric mean inequality. Let $a, b \in \mathbb{R}$. Prove that for any $\varepsilon > 0$

$$ab \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2. \quad (2.32)$$

Hint: expand $(\sqrt{\varepsilon} a \pm b/\sqrt{\varepsilon})^2 \geq 0$ and rearrange.

$$\left(\sqrt{\varepsilon} a - \frac{b}{\sqrt{\varepsilon}} \right)^2 \geq 0 \Rightarrow \varepsilon a^2 + \frac{b^2}{\varepsilon} - 2ab \geq 0$$

$$\Rightarrow ab \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2$$

11. (Optional for undergrads) Let Ω be bounded and suppose that $f_n \rightarrow f$ in $L^p(\Omega)$, i.e. $\|f_n - f\|_{L^p(\Omega)} \rightarrow 0$. Prove that

$$\int_{\Omega} f_n(x) dx \xrightarrow{n \rightarrow \infty} \int_{\Omega} f(x) dx.$$

Hint: estimate $\left| \int_{\Omega} (f_n - f) dx \right| \leq \int_{\Omega} |f_n - f| dx = \|f_n - f\|_{L^1(\Omega)}$ and then use the previous exercise.

$$\left| \int_{\Omega} (f_n - f) dx \right| \leq \int_{\Omega} |f_n - f| dx = \|f_n - f\|_{L^1(\Omega)} \leq |\Omega|^{\frac{1}{p} - \frac{1}{q}} \|f_n - f\|_{L^p(\Omega)}$$

As Ω is bounded,
 $|\Omega|^{\frac{1}{p} - \frac{1}{q}} \|f_n - f\|_{L^p(\Omega)} \rightarrow 0$

$$\Rightarrow \left| \int_{\Omega} (f_n - f) dx \right| \rightarrow 0$$

12. Let $\psi \in C^1(\Omega)$ and let $\vec{\alpha}$ be a multi-index with $|\vec{\alpha}| = 1$, say $\vec{\alpha} = (0, \dots, 1, \dots, 0)$ with the 1 in the i -th position. Prove that the weak derivative $D^{\vec{\alpha}} \psi$ of Definition 2.4.1 exists and coincides with the classical partial derivative $\partial \psi / \partial x_i$. Hint: start from $\int_{\Omega} \frac{\partial \psi}{\partial x_i} \phi dx$ and integrate by parts. Since $\phi \in \mathcal{D}(\Omega)$ vanishes in a neighbourhood of $\partial \Omega$, the boundary term $\int_{\partial \Omega} \psi \phi n_i ds$ is zero, and the identity of Eq. (2.24) follows at once. (Graduate: extend the result to an arbitrary multi-index $\vec{\alpha}$ and $\psi \in C^{|\vec{\alpha}|}(\Omega)$, by induction on $|\vec{\alpha}|$; each integration by parts produces one factor (-1) , which explains the sign $(-1)^{|\vec{\alpha}|}$ in Eq. (2.25).)

Definition 2.4.1 (Weak Partial Derivative) A locally integrable function $f: \Omega \rightarrow \mathbb{R}$ has a weak derivative $D^i f$ with respect to x_i if there exists g satisfying

$$\int_{\Omega} g(x) \phi(x) dx = - \int_{\Omega} f(x) \frac{\partial \phi}{\partial x_i}(x) dx \quad \text{for all } \phi \in \mathcal{D}(\Omega). \quad (2.24)$$

We then set $D^i f = \frac{\partial f}{\partial x_i} := g$.

$$\frac{\partial}{\partial x_i} (\gamma \phi) = \frac{\partial \gamma}{\partial x_i} \phi + \gamma \frac{\partial \phi}{\partial x_i} \Rightarrow \frac{\partial \gamma}{\partial x_i} \phi = \frac{\partial}{\partial x_i} (\gamma \phi) - \gamma \frac{\partial \phi}{\partial x_i}$$

$$\int_{\Omega} \frac{\partial \gamma}{\partial x_i} \phi dx = \int_{\Omega} \frac{\partial}{\partial x_i} (\gamma \phi) dx - \int_{\Omega} \gamma \frac{\partial \phi}{\partial x_i} dx$$

$$\begin{aligned} \frac{\partial}{\partial x_i} (\gamma \phi) &= \nabla \cdot (\gamma \phi \vec{e}_i) \Rightarrow \int_{\Omega} \frac{\partial}{\partial x_i} (\gamma \phi) dx = \int_{\partial \Omega} \gamma \phi \vec{e}_i \cdot \vec{n} ds = \\ &= \int_{\partial \Omega} \gamma \phi n_i ds = 0 \\ \Rightarrow \int_{\Omega} \frac{\partial \gamma}{\partial x_i} \phi dx &= - \int_{\Omega} \gamma \frac{\partial \phi}{\partial x_i} dx \end{aligned}$$

Base case: $|\vec{\alpha}|=1$

Already proven

$$|\vec{\alpha}|=n \Rightarrow |\vec{\alpha}'|=n+1$$

Suppose $\vec{\alpha}=(\alpha_1, \dots, \alpha_d)$, $|\vec{\alpha}'|=n$. We have

$$\int_{\Omega} D^{\vec{\alpha}} \gamma \varphi dx = (-1)^n \int_{\Omega} \gamma D^{\vec{\alpha}} \varphi dx$$

Take $\vec{\alpha}'_2 = (\alpha_1, \dots, \alpha_i+1, \dots, \alpha_d)$. $|\vec{\alpha}'_2|=n+1$

$$\int_{\Omega} D^{\vec{\alpha}'_2} \gamma \varphi dx = \int_{\Omega} \frac{\partial}{\partial x_i} (D^{\vec{\alpha}} \gamma) \varphi dx =$$

$$= \int_{\Omega} \frac{\partial}{\partial x_i} (D^{\vec{\alpha}} \gamma \varphi) dx - \int_{\Omega} D^{\vec{\alpha}} \gamma \frac{\partial \varphi}{\partial x_i} dx$$

$$\frac{\partial}{\partial x_i} (D^{\vec{\alpha}} \gamma \varphi) = \nabla \cdot \left((0, \dots, \underset{i\text{-th}}{1}, \dots, 0) D^{\vec{\alpha}} \varphi \right)$$

$$\Rightarrow \int_{\Omega} \frac{\partial}{\partial x_i} (D^{\vec{\alpha}} \gamma \varphi) dx = \int_{\partial \Omega} D^{\vec{\alpha}} \gamma \varphi n_i ds = 0$$

Note that $\frac{\partial \varphi}{\partial x_i}$ is also a test function $\Rightarrow$

$$\int_{\Omega} D^{\vec{\alpha}} \gamma \frac{\partial \varphi}{\partial x_i} dx \stackrel{IH}{=} (-1)^n \int_{\Omega} \gamma D^{\vec{\alpha}} \left(\frac{\partial \varphi}{\partial x_i} \right) dx =$$

$$= (-1)^n \int_{\Omega} \gamma D^{\vec{\alpha}'_2} \varphi dx$$

$$\Rightarrow \int_{\Omega} D^{\vec{\alpha}'_2} \gamma \varphi dx = (-1)^{n+1} \int_{\Omega} \gamma D^{\vec{\alpha}'_2} \varphi dx =$$

$$= (-1)^{n+1} \int_{\Omega} \gamma D^{\vec{\alpha}} \varphi dx$$